ENGINEERING CHANGE RECORD

DragonBox Pyra Kernel

OMAP5432 Cortex-M4 Remoteproc Port & Stability Fixes

Complete project change summary | Baseline: 5.6.19-pyradef / Debian packaging revision -10

Executive summary

This report documents the modifications introduced by the Pyra M4 kernel project, relative to the original Pyra 5.6.19-pyradef source tree with its Debian -10 packaging diff applied. It covers the OMAP5 remoteproc driver import, M4-specific Device Tree and kernel configuration, build compatibility adjustments, Phase 9.1 and 9.2 stability/USB backports, and the newly applied Phase 9.3 OMAP PRM reset fix.

Item	Current assessment
Kernel release	5.6.19-pyradef-m4test (existing build; 9.3 rebuild not yet confirmed)
M4 registration	Confirmed: remoteproc0 = 55020000.ipu, initially offline
PowerVR compatibility	Confirmed at module level: SGX DDK user/kernel 3699939 match
M4 execution	Not confirmed; v3 encountered IPU IOMMU fault at DA 0x0
Phase 9.3	Patch applied to sources per user; rebuilt/boot-tested image not yet verified

Scope note. “All changes” below means changes made by this project to the supplied Pyra kernel baseline. Existing board-specific drivers and the Debian packaging diff are retained, not attributed to this project.

1 Baseline and architecture

Component	Baseline / objective
Platform	DragonBox Pyra; Texas Instruments OMAP5432 (dual ARM Cortex-A15 + IPU with two Cortex-M4 cores).
Source	linux-5.6.19-pyradef_5.6.19-pyradef.orig.tar.gz plus linux-5.6.19-pyradef_5.6.19-pyradef-10.diff.gz.
Build environment	ARMv7 / Debian 12, GCC 12; out-of-tree Linux Kbuild using a separate build directory.
Graphics constraint	Keep original PowerVR SGX544MP2 1.14.3699939 kernel interface and Pyra display components untouched.
Boot isolation	Original Pyra V5.3 DTB stays available; an additional m4test DTB enables the IPU.

2 Change map — at a glance

The project touches 14 kernel source paths: nine for M4 enablement, configuration and compiler compatibility; three for Phase 9.1 (charger, MMC and TILER); one for Phase 9.2 (DWC3/OTG); and one for Phase 9.3 (PRM reset). Build and diagnostic scripts are external tooling and do not modify the running kernel by themselves.

3 Cortex-M4 integration and build adjustments

3.1 Linux 5.7 OMAP remoteproc backport

The original drivers/remoteproc/omap_remoteproc.c and omap_remoteproc.h are replaced with Device Tree-aware Linux v5.7 equivalents, adapted to Linux 5.6.19 interfaces. Prior local versions are preserved as *.pre-m4 backups. The imported driver matches ti,omap5-ipu and creates one remoteproc object for the IPU subsystem, not separate Linux remoteproc devices for each Cortex-M4 core.

Implementation detail	Modification
Timer compatibility	set_load(timer->odt, 0) → set_load(timer->odt, 0, 0).
Remoteproc API	da_to_va(..., size_t len) → da_to_va(..., int len).
Boot-address format	Format 32-bit address with 0x%08x instead of 0x%llx.
Safe startup	rproc->auto_boot = false before rproc_add(); no automatic M4 firmware execution.
Internal memory	IPU L2 RAM is mapped for remoteproc via the driver's device-address translation.

3.2 Optional IPU Device Tree

A disabled ti,omap5-ipu node is added to arch/arm/boot/dts/omap5.dtsi with physical L2-SRAM registers at 0x55020000 (64 KiB), IPU IOMMU, reset lines, clocks, mailbox and firmware-name. A new Pyra V5.3 -m4test.dts includes the existing board DTS and overrides only the IPU node to status = "okay". The original board DTS remains unchanged.

```
Default OMAP5 node:      /ocp/ipu@55020000  status = "disabled"
Additional Pyra DTB:     omap5-letux-cortex15-v5.3+pyra-v5.3-m4test.dtb
M4-enabled override:     &ipu { status = "okay"; };
Firmware default name:   omap5-ipu-fw.xem4
```

There is no dedicated /reserved-memory node or memory-region property in the current IPU DT definition. The kernel therefore warns that the IPU has no specific CMA pool. This is intentional for the initial registration phase; it is not a validated DDR-backed firmware configuration.

3.3 Kernel configuration and host/compiler compatibility

- pyra_defconfig: LOCALVERSION changes to -pyradef-m4test; CONFIG_REMOTEPROC=y, CONFIG_OMAP_REMOTEPROC=y and CONFIG_OMAP2PLUS_MBOX=y are enabled. The build script additionally verifies CONFIG_OMAP_IOMMU=y and relevant mailbox options.
- arch/arm/Makefile: move -msoft-float early into KBUILD_CFLAGS so GCC 12 capability probes also use kernel-appropriate floating-point flags, rather than appending it only after earlier probes.
- security/selinux/include/classmap.h: guard the PF_MAX consistency check with ! defined(__EXPORTED_HEADERS__) to avoid host-header compatibility errors.
- drivers/remoteproc/Kconfig: update the OMAP remoteproc help text to describe OMAP4 and OMAP5 Device Tree support.

4 Phase 9.1 — targeted stability corrections

File	Exact behavioral change / rationale
drivers/power/supply/bq2429x_charger.c	Remove two stray semicolons following if (!of_property_read_u32(...)); optional precharge-current and charge-termination-current settings are applied only when their DT properties are present. No battery limits or DT settings are changed.
drivers/mmc/host/omap_hsmmc.c	Calculate timeout as (u64)req->cmd->busy_timeout * NSEC_PER_MSEC to avoid 32-bit overflow for MMC busy commands.
drivers/gpu/drm/omapdrm/omap_dmm_tiler.c	On refill-buffer allocation failure, set ret = -ENOMEM before the existing failure cleanup path. No GPU/TILER throughput or memory policy change.

5 Phase 9.2 — USB DWC3 OMAP corrections

All changes are confined to drivers/usb/dwc3/dwc3-omap.c; the DWC3 driver is not replaced wholesale.

Area	Change
VBUS disable	Only call regulator_disable() when vbus_reg exists and regulator_is_enabled() reports an enabled regulator.
Initial extcon state	Explicitly set OMAP_DWC3_VBUS_OFF when USB VBUS is absent and OMAP_DWC3_ID_FLOAT when host ID is not grounded; previously only asserted states were initialized.
Optional VBUS regulator	Use devm_regulator_get_optional(dev, "vbus") instead of a vbus-supply property pre-check; treat -ENODEV as absent, propagate other errors including probe deferral.
IRQ probe failure	After an IRQ request error, call of_platform_depopulate(dev) before unwinding runtime power management, preventing orphaned child devices.

6 Phase 9.3 — OMAP PRM atomic reset fix

The Phase 9.3 patch targets drivers/soc/ti/omap_prm.c, in omap_reset_deassert(). It applies upstream Linux commit 98ece19f247159a51003796ede7112fef2df5d7f, attributed in the supplied patch to Tero Kristo (TI).

Before	After
readl_relaxed_poll_timeout(...)	readl_relaxed_poll_timeout_atomic(...)

Reason: the old polling macro can sleep via usleep_range(), but this reset path can execute in an atomic context during IOMMU runtime-PM and remoteproc startup. The observed firmware-v3 logs contain “BUG: scheduling while atomic” with omap_reset_deassert() in the call chain. The fix changes reset polling only; no reset timing constant, clock tree, DT property or GPU code is modified.

Verification boundary. The user confirmed applying this patch to the kernel sources. There is not yet an independently verified successful build, installation, boot or v3 firmware retest of the Phase 9.3 image.

Appendix A — file-by-file change inventory

Kernel source path	Change set
arch/arm/Makefile	M4 / build
security/selinux/include/classmap.h	M4 / build
arch/arm/boot/dts/omap5.dtsi	M4 / Device Tree
arch/arm/boot/dts/omap5-letux-cortex15-v5.3+pyra-v5.3-m4test.dts	M4 / new file
arch/arm/boot/dts/Makefile	M4 / Device Tree
arch/arm/configs/pyra_defconfig	M4 / configuration
drivers/remoteproc/Kconfig	M4 / configuration
drivers/remoteproc/omap_remoteproc.c	M4 / Linux 5.7 import
drivers/remoteproc/omap_remoteproc.h	M4 / Linux 5.7 import
drivers/power/supply/bq2429x_charger.c	Phase 9.1
drivers/mmc/host/omap_hsmmc.c	Phase 9.1
drivers/gpu/drm/omapdrm/omap_dmm_tiler.c	Phase 9.1
drivers/usb/dwc3/dwc3-omap.c	Phase 9.2
drivers/soc/ti/omap_prm.c	Phase 9.3

Appendix B — source provenance / audit trail

The following locally supplied project artifacts were inspected to prepare this document:

- Pyra-Kernel-M4-5.6.19-pyradef-10-v4.zip — 0001-pyra-m4-kernel-and-dt.patch; 0002-pyra-phase-9.1-9.2-stability-usb.patch; five setup/build/diagnostic scripts; README.md.
- pyra-m4-omap-prm-atomic-reset-fix.patch — independent Phase 9.3 kernel patch (upstream commit 98ece19f247159a51003796ede7112fef2df5d7f).
- pyra-m4-v3-dmesg.log; pyra-m4-v3-status.log; pyra-m4-v3-first-run.log — hardware diagnosis for the v3 M4 firmware.
- User-supplied running-kernel command output: uname, lsmod, modinfo, remoteproc state, Device Tree, memory map and PowerVR DDK match.